ELEC4631 Solutions Tutorial 3

1. (a) f is positive in the interval $(-5,5)$ and is negative for $x < -5$ and $x > 5$. In the region $(-5,5)$ the function is zero only at x^*. Therefore f is lpd.

   (b) f is an even function $f(-x) = f(x)$ so it is enough to look at f for $x \geq 0$. By L'Hôpital's rule from calculus,

   $$\lim_{x\to 0} \frac{\sin(x)}{x} = \lim_{x\to 0} \frac{\cos(x)}{1} = 1.$$

   Therefore, f is a continuous function since $\lim_{x\to 0} f(x) = 1 - \lim_{x\to 0} \frac{\sin(x)}{x} = 0 = f(0)$.

   Let $g(x) = x - \sin(x)$. Then we can write $f(x) = \frac{g(x)}{x}$. We have that for $x \geq 0$,

   $$g(x) = g(0) + \int_0^x \frac{dg(v)}{dx} dv = \int_0^x (1 - \cos(v))dv.$$

   Since $1 - \cos(v) \geq 0$ and it is not the zero function, $\int_0^x (1-\cos(v))dv > 0$ when $x > 0$. Therefore $g(x) > 0$ for $x > 0$ and also $f(x) > 0$ for all $x > 0$. In conclusion, f is a positive definite function.

   (c) Clearly, $f(x,y,z) \leq 0$ for all x, y, z. Furthermore, $f(x,y,z) = 0$ implies that $(x-2)^2 = 0$, $(y+1)^4 = 0$ and $(2x-z)^2 = 0$ thus $x = 2$, $y = -1$ and $z = 2x = 4$. That is, f is negative everywhere except at $(x,y,z) = (2,-1,4)$. Thus, f is nd around $(2,-1,4)$.

   (d) Clearly $f(x,y,z) = -(y+1)^4 - (2x-z)^2 \leq 0$ for all (x,y,z) and $f(2,-1,4) = 0$. So, f is nsd around $(2,-1,4)$ but there's more. Now, $f(x,y,z) = 0$ implies that $(y+1)^2 = 0$ and $(2x-z)^2 = 0$ thus $y = -1$ and $z = 2x$. That is f vanishes at all points of the form $(x,-1,2x)$. It follows that for any $r > 0$, the region $0 < \|(x,y,z) - (2,-1,4)\| < r$ always contains a point different from $(2,-1,4)$ at which f vanishes, e.g., all points $(a,-1,2a)$ for any $a \neq 2$ satisfying $(a-2)^2 + (2a-4)^2 < r^2$. That is, f is lnsd around $(2,-1,4)$.

   In conclusion, f is both nsd and lnsd around $(2,-1,4)$.

2. (a) The vector field is $f(x) = \begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix} x$. Since an equilibrium point x^* satisfies $f(x^*) = 0$, the equilibrium points for this system are all points in null $\left(\begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix} \right)$.

(b) Method 1: Write $V(x) = x^T \begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix} x$

$$\begin{aligned} \dot{V}(x) &= \dot{x}^T \begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix} x + x^T \begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix} \dot{x}, \\ &= x^T \begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix}^T \begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix} x + x^T \begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix} \begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix} x, \\ &= x^T \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} x. \end{aligned}$$

Method 2: Write $\nabla V(x) = (10x_1 - 14x_2, -14x_1 + 20x_2)^T$ so

$$\begin{aligned} \dot{V}(x) &= \nabla V(x)^T f(x), \\ &= \begin{bmatrix} 10x_1 - 14x_2 & -14x_1 + 20x_2 \end{bmatrix} \begin{bmatrix} -3x_1 + 3x_2 \\ -2x_1 + 2x_2 \end{bmatrix}, \\ &= -2x_1^2 + 4x_1x_2 - 2x_2^2, \\ &= x^T \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} x. \end{aligned}$$

(c) $\begin{bmatrix} 5 & -7 \\ -7 & 10 \end{bmatrix}$ has two positive eigenvalues 0.067 and 14.933. Therefore, V is a positive definite quadratic form, in particular $V(x) = 0 \Leftrightarrow x = 0$. So, V is also lpd around the origin.

(d) $\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ has eigenvalues -4 and 0, so V is a negative semidefinite quadratic form. Negative semidefinite quadratic forms are automatically lnsd, and they cannot be lnd. This is because there is a line formed by null $\left(\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \right)$ at which $\dot{V}$ is zero. **It is important to keep in mind, however, that this is not true for general negative semidefinite functions (that are not quadratic forms). Negative semidefiniteness can imply lnsd or lnd**.

(e) V is lpd and $\dot{V}$ lnsd around the origin from (b) and (c), so the origin is stable i.s.L.

Remark: It is clear that V is in fact a global Lyapunov function around the origin. But since we only want to show stability i.s.L, the global property is not really needed and only the local property is mentioned above

3. (a) $V(x, y)$ is a positive definite function (not only locally) around the origin.

$$\begin{aligned} \dot{V}(x, y) &= 2x\dot{x} + 2y\dot{y}, \\ &= 2x(-x + y - xy^2) + 2y(-x - y - x^2 y), \\ &= -2(x^2 + 2x^2y^2 + y^2). \end{aligned}$$

Since $x^2 \geq 0$, $x^2y^2 \geq 0$ and $y^2 \geq 0$, $\dot{V}(x, y) = 0 \Leftrightarrow x = 0,\ y = 0$. While if $(x, y) \neq (0, 0)$, $\dot{V}(x, y) < 0$. Thus V is a negative definite function (not only locally) around the origin. Since V is also radially unbounded, from Lyapunov's theorem the origin is globally asymptotically stable.

(b) $V(x, y)$ is pd around the origin as in previous sub-problem.

$$\begin{aligned} \dot{V}(x, y) &= 2x\dot{x} + 2y\dot{y}, \\ &= 2x(-x^3 + y^4) + 2y(-y^3(x + 1)), \\ &= -2(x^4 + y^4). \end{aligned}$$

By same kind of argument as in previous sub-problem, $\dot{V}(x, y) = 0 \Leftrightarrow x = 0,\ y = 0$. While if $(x, y) \neq (0, 0)$, $\dot{V}(x, y) < 0$. Thus V is a positive definite function around the origin. Since V is also radially unbounded, from Lyapunov's theorem the origin is globally asymptotically stable.

(c) $V(x, y)$ is pd around the origin using same arguments as in previous sub-problems.

$$\begin{aligned} \dot{V}(x, y) &= 2x\dot{x} + 2y\dot{y}, \\ &= 2x(-x + 4y) + 8y(-x - y^3), \\ &= -2(x^2 + 4y^4). \end{aligned}$$

By same kind of argument as in previous sub-problems, $\dot{V}(x, y) = 0 \Leftrightarrow x = 0,\ y = 0$. While if $(x, y) \neq (0, 0)$, $\dot{V}(x, y) < 0$. Thus V is a positive definite function around the origin. Since V is also radially unbounded, from Lyapunov's theorem the origin is globally asymptotically stable.

4. (a) By completion of squares, write $V(v_C, i)$ as

$$V(v_C, i) = \frac{1}{2}(v_C + \epsilon L i)^2 + \frac{1}{2}\left(\frac{L}{C} - \epsilon^2 L^2\right) i^2.$$

Since $\epsilon > 0$ can be chosen as small as we want (but non-zero), we choose it such that $\epsilon < \frac{1}{\sqrt{LC}}$. Choosing $\epsilon > 0$ this way guarantees $V(v_C, i) = 0$ only if $v_C = 0$ an $i = 0$, at all other points the function is positive. Therefore, $V(v_C, i)$ becomes a positive definite quadratic form.
Repeat the same for $\dot{V}(v_C, i)$:

$$\begin{aligned}
\dot{V}(v_C, i) &= \nabla V(v_C, i)^T f(v_C, i), \\
&= \frac{\epsilon L}{C} i^2 - \frac{R}{C} i^2 - \epsilon R v_C i - \epsilon v_C^2, \\
&= -\frac{1}{C}(R - \epsilon L) i^2 - \epsilon R v_C i - \epsilon v_C^2, \\
&= -\epsilon (v_C + \frac{R}{2} i)^2 + \frac{\epsilon R^2}{4} i^2 - \frac{1}{C}(R - \epsilon L) i^2, \\
&= -\left(\epsilon (v_C + \frac{R}{2} i)^2 + \frac{1}{C}(R - \epsilon(L + \frac{R^2 C}{4})) i^2 \right),
\end{aligned}$$

and choosing $0 < \epsilon < \frac{R}{\frac{R^2C}{4} + L}$ ensures that $R - \epsilon(L + \frac{R^2C}{4}) > 0$. It follows that $\dot{V}$ will be a negative definite quadratic form.
By choosing $0 < \epsilon < \min(\frac{1}{\sqrt{LC}}, \frac{R}{\frac{R^2C}{4} + L})$ we *simultaneously* make V a positive definite quadratic form and $\dot{V}$ a negative definite quadratic form.

(b) Since V is a positive definite quadratic form, (i) it is positive everywhere except at (0,0). Being a positive definite quadratic form $V(x)$ is radially unbounded as a function of x ((note that a positive semidefinite function need not be radially unbounded). Finally, (iii) $\dot{V}$ is a negative definite quadratic form thus it is negative everywhere except at (0,0). Therefore, from (i)-(iii) $(0,0)$ is a globally asymptotically stable equilibrium point.

5. (a) Since (i) $\omega^2 \geq 0$, $\left(\frac{1}{D}\omega + \theta\right)^2 \geq 0$ and $1 - \cos\theta \geq 0$ then (ii) $V(\theta, \omega) = 0 \Leftrightarrow \omega^2 = 0$, $\left(\frac{1}{D}\omega + \theta\right)^2$, $1 - \cos\theta = 0 \Leftrightarrow \theta = 0$, $\omega = 0$. From (i) and (ii), V is lpd around the origin (in fact it is more, it is a positive definite function).

(b)

$$\begin{aligned}
\dot{V}(x) &= \nabla V(x)^T f(x), \\
&= \left[\ \tfrac{\omega}{D} + \theta + \tfrac{g}{r}(1+\tfrac{1}{D^2})\sin\theta \quad (1+\tfrac{1}{D^2})\omega + \tfrac{\theta}{D}\ \right] \left[\begin{array}{c} \omega \\ -D\omega - \frac{g}{r}\sin\theta \end{array} \right], \\
&= \frac{\omega^2}{D} + \omega\theta + \frac{g}{r}(1+\frac{1}{D^2})\omega\sin\theta - D\omega^2 - \frac{1}{D}\omega^2 - \frac{g}{r}(1+\frac{1}{D^2})\omega\sin\theta, \\
&\quad -\theta\omega - \frac{g}{rD}\theta\sin\theta, \\
&= -(D\omega^2 + \frac{g}{rD}\theta\sin\theta).
\end{aligned}$$

(c) $g'(\theta) = \theta\cos\theta + \sin\theta \to g'(0) = 0$, $g''(\theta) = 2\cos\theta - \theta\sin\theta \to g''(0) > 0$. So, 0 is a local minimum of g and there is a constant $r > 0$ such that $g(\theta) > g(0)$ for all $0 < |\theta| < r$. Hence, g is lpd around 0.

6. The vector field F on $[-\delta, \delta] \times [-\delta, \delta]$ here is

$$F(q,p) = \left[\begin{array}{c} p \\ -p^3 - \sin(q)^5 \end{array} \right].$$

(a) Setting $F(q, p) = 0$ to find the equilibrium points, we find that $p = 0$ and $q = 0$. We conclude that $(0, 0)$ is the only equilibrium point in the region $[-\delta, \delta] \times [-\delta, \delta]$.

(b) Let $V_1(q) = \int_0^q \sin(u)^5 du$. First note that $V_1(0) = 0$. When $0 < q \le \delta < \pi/2$ we have that $\sin(q)^5 > 0$. Therefore also $V_1(q) = \int_0^q \sin(u)^5 du > 0$ when $0 < q \le \delta$. When $-\pi/2 < -\delta \le q < 0$ we have that $\sin(q)^5 < 0$. However, $\int_0^q \sin(u)^5 du = -\int_q^0 \sin(u)^5 du = \int_q^0 (-\sin(u)^5) du > 0$. Therefore we conclude that the integral $V_1(q) = \int_0^q \sin(u)^5 du > 0$ when $-\pi/2 < -\delta \le q < 0$. In summary, $V_1(0) = 0$ and $V_1(q) > 0$ when $0 < |q| \le \delta < \pi/2$.

Since $V(q,p) = V_1(q) + \frac{1}{2}p^2$, $V(q, p) \ge 0$ for all $|q| \le \delta$ and all $|p| \le \delta$, and $V(q, p) = 0$ on $[-\delta, \delta] \times [-\delta, \delta]$ implies that $V_1(q) = 0$ and $\frac{1}{2}p^2$, hence $q = 0$ and $p = 0$. Therefore, V is lpd around the origin.

(c)

$$\begin{aligned}
\dot{V}(q,p) &= \nabla V(q,p)^T F(q,p), \\
&= \left[\ \sin(q)^5 \quad p\ \right] \left[\begin{array}{c} p \\ -p^3 - \sin(q)^5 \end{array} \right], \\
&= -p^4.
\end{aligned}$$

So, $\dot{V}(q,p) < 0$ for all $0 < |p| \leq \delta$ but $\dot{V}(q,p) = 0$ whenever $p = 0$ for any $|q| \leq \delta$. We can only conclude that $\dot{V}$ is lnsd but not lnd in the region $-\delta \leq q \leq \delta$ and $-\delta \leq p \leq \delta$. It is not lnd because $\dot{V}(q,0) = 0$ for any $0 < |q| \leq \delta$ and not only at $q = 0$. Since V is lpd from part (b) and $\dot{V}$ is lnsd around $(0,0)$, we conclude that V is a Lyapunov function around $(0,0)$. So, it follows that $(0,0)$ is stable i.s.L.

(d) To apply the Krasovskii-LaSalle invariance principle, we look for trajectories that satisfy $\dot{V}(q(t),p(t)) = 0$ for all times $t \geq 0$. That is, trajectories for which $p(t) = 0$ for all times $t \geq 0$. If $p(0) = 0$ and $p(t) = 0$ for all $t > 0$, we also have $\dot{p}(t) = 0$ for all $t \geq 0$. Since

$$\dot{p}(t) = -f(p(t) - g(q(t)).$$

From $\dot{p}(t) = 0$ and $p(t) = 0\ \forall t \geq 0$ we get that $g(q(t)) = 0\ \forall t \geq 0$. In the region $[-\delta,\delta] \times [-\delta,\delta]$, $g(q(t)) = 0$ implies that $q(t) = 0$. Thus the only trajectory for which $\dot{V}(q(t),p(t)) = 0$ for all times $t \geq 0$ is the constant trajectory $p(t) = 0$ and $q(t) = 0$. The largest positively invariant set M in the set I in the Krasovskii-LaSalle invariance principle contains only the origin. Therefore, the origin is asymptotically stable.